

EXP NO:5

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

CSV FILE

SepalLength,SepalWidth,PetalLength,PetalWidth,Species

5.1,3.5,1.4,0.2,Setosa

4.9,3.0,1.4,0.2,Setosa

5.0,3.4,1.5,0.2,Setosa

6.4,3.2,4.5,1.5,Versicolor

6.9,3.1,4.9,1.5,Versicolor

5.5,2.3,4.0,1.3,Versicolor

6.5,3.0,5.8,2.2,Virginica

7.6,3.0,6.6,2.1,Virginica

7.3,2.9,6.3,1.8,Virginica

PROGRAM

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = pd.read_csv("iris.csv")

print(data)

X = data.iloc[:, :-1]
y = data.iloc[:, -1]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.3,  
    random_state=42  
)
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train, y_train)
```

```
y_pred = knn.predict(X_test)
```

```
print("\nActual Classes:")  
print(y_test.values)
```

```
print("\nPredicted Classes:")  
print(y_pred)
```

```
accuracy = accuracy_score(y_test, y_pred)  
print("\nAccuracy:", accuracy)
```

```
new_sample = pd.DataFrame(  
    [[5.9, 3.0, 5.1, 1.8]],  
    columns=X.columns  
)
```

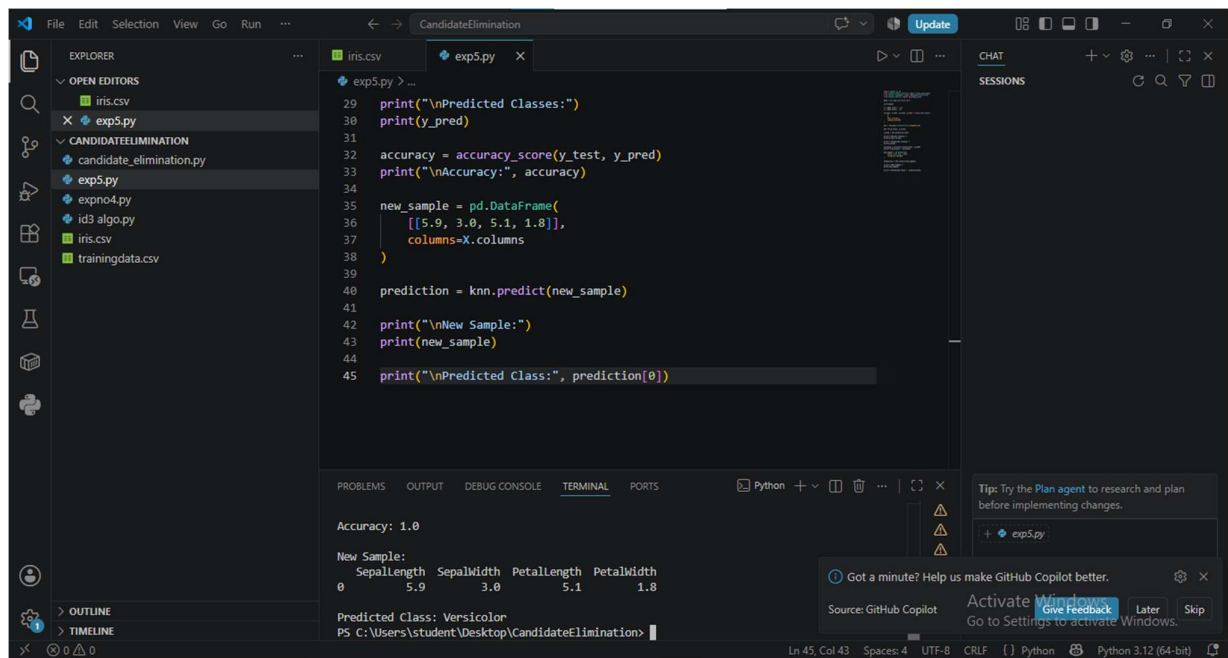
```
prediction = knn.predict(new_sample)
```

```
print("\nNew Sample:")
```

```
print(new_sample)
```

```
print("\nPredicted Class:", prediction[0])
```

OUTPUT



```
File Edit Selection View Go Run ... CandidateElimination Update
EXPLORER
  OPEN EDITORS
    iris.csv
    exp5.py
  CANDIDATEELIMINATION
    candidate_elimination.py
    exp5.py
    expno4.py
    id3_algo.py
    iris.csv
    trainingdata.csv
  PROBLEMS
  OUTPUT
  DEBUG CONSOLE
  TERMINAL
  PORTS
  Python + - [] ... | < >
  Python 3.12 (64-bit)

29 print("\nPredicted Classes:")
30 print(y_pred)
31
32 accuracy = accuracy_score(y_test, y_pred)
33 print("\nAccuracy:", accuracy)
34
35 new_sample = pd.DataFrame(
36     [[5.9, 3.0, 5.1, 1.8]],
37     columns=X.columns
38 )
39
40 prediction = knn.predict(new_sample)
41
42 print("\nNew Sample:")
43 print(new_sample)
44
45 print("\nPredicted Class:", prediction[0])

Accuracy: 1.0

New Sample:
   SepalLength  SepalWidth  PetalLength  PetalWidth
0           5.9          3.0          5.1          1.8

Predicted Class: Versicolor
PS C:\Users\student\Desktop\CandidateElimination>
```

Tip: Try the [Plan agent](#) to research and plan before implementing changes.

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